

12.206

AI25BTECH11024 - Pratyush Panda

Question:

The eigenvectors of the matrix

$$\begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \quad (0.1)$$

are written in the form $\begin{pmatrix} 1 \\ a \end{pmatrix}$ and $\begin{pmatrix} 1 \\ b \end{pmatrix}$. What is $a + b$?

- a 0
- b $\frac{1}{2}$
- c 1
- d 2

Solution:

Given

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \quad (0.2)$$

We have to find the eigen vectors. For that we first have to find the eigen values using the characteristic equation

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0 \quad (0.3)$$

$$(1 - \lambda)(2 - \lambda) = 0 \quad (0.4)$$

$$(0.5)$$

Hence,

$$\lambda_1 = 1, \quad \lambda_2 = 2 \quad (0.6)$$

Now, to find the eigenvector for each eigen value;

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad (0.7)$$

$$\implies (\mathbf{A} - \lambda \mathbf{I})\mathbf{v} = 0 \quad (0.8)$$

$$\begin{pmatrix} 1 - \lambda & 2 \\ 0 & 2 - \lambda \end{pmatrix} \mathbf{v} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (0.9)$$

Writing as an augmented matrix and row-reducing:

$$\left(\begin{array}{cc|c} 1 - \lambda & 2 & 0 \\ 0 & 2 - \lambda & 0 \end{array} \right) \quad (0.10)$$

For $\lambda = 1$ we have;

$$\left(\begin{array}{cc|c} 0 & 2 & 0 \\ 0 & 1 & 0 \end{array} \right) \quad (0.11)$$

On solving, we get $\mathbf{v} = \begin{pmatrix} x \\ 0 \end{pmatrix}$.

Thus, the first eigen vector is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

For $\lambda = 2$ we have;

$$\left(\begin{array}{cc|c} -1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad (0.12)$$

On solving this, we get $\mathbf{v} = \begin{pmatrix} x \\ \frac{x}{2} \end{pmatrix}$.

Thus, the second eigen vector is $\begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix}$.

Thus, when we get $a = 0$ and $b = \frac{1}{2}$. So $a + b = \frac{1}{2}$. Therefore, option d is correct.